

PROJECT REPORT

❖ **Project Context:**

Out of the six project choices provided, I selected the following:

- **Project Name:** Retail Store Customer Clustering
- **Skills Covered:** K-Means clustering, PCA, Seaborn
- **Brief Description:** Apply unsupervised learning to segment customers based on purchase patterns and visualize with PCA.

This project was chosen because it provides hands-on experience with clustering techniques, dimensionality reduction, and data visualization, all of which are essential skills in Data Science.

❖ **Introduction:**

- Retail businesses handle large volumes of customer data such as age, gender, annual income, and spending patterns. Understanding customer behaviour is essential for targeted marketing, personalized services, and better decision-making.
- In this project, I have applied **unsupervised learning (K-Means clustering)** along with **Principal Component Analysis (PCA)** to group customers into meaningful segments based on their demographic and spending characteristics. Visualization techniques help us interpret clusters effectively and derive actionable insights.

❖ **Objective:**

The main objectives of this project are:

- To apply **K-Means clustering** on mall customers data to identify different customer segments.
- To use **PCA** for dimensionality reduction and visualization of high-dimensional data in 2D space.
- To interpret and profile the clusters in terms of age, income, and spending score.
- To provide insights that can assist businesses in designing marketing strategies and customer retention plans.

❖ Methodology:

➤ Tools and Technologies:

- **Python** programming language.
- **Google Colab** as the development environment.
- Libraries used: pandas, numpy, matplotlib, seaborn, scikit-learn, joblib.

➤ Steps Followed:

1. **Dataset Loading:** The *Mall Customers Dataset* from Kaggle was used.
2. **Data Cleaning & Preprocessing:** Columns were standardized, missing values checked, and categorical variables explored.
3. **Exploratory Data Analysis (EDA):** Histograms, boxplots, and heatmaps were created to study distributions and correlations.
4. **Feature Selection:** Age, Annual Income, and Spending Score were selected as key features.
5. **Data Scaling:** StandardScaler was applied to normalize data.
6. **Dimensionality Reduction:** PCA was applied to reduce dimensions to 2 for visualization while preserving maximum variance.
7. **Clustering:** K-Means clustering was used with the optimal number of clusters determined using the **Elbow Method** and **Silhouette Score**.
8. **Visualization:** Clusters were plotted in original feature space and PCA-reduced 2D space.
9. **Profiling & Labeling:** Cluster centroids were analyzed to assign meaningful labels (e.g., “High Income – Low Spend”).
10. **Saving Results:** Final datasets with cluster labels, centroids, and trained models were exported for future use.

❖ Code and Implementation Details:

The implementation was carried out in **14 well-structured steps**. Below, are the key parts of the code along with an explanation of their functionality.

➤ Step 1 – Colab Setup & Dataset Upload:

```
from google.colab import files
uploaded = files.upload()
```

- This step allows us to upload the dataset into Google Colab.
- Optionally, Google Drive can be mounted for persistent storage.

➤ **Step 2 – Import Libraries and Load Dataset:**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import joblib

df = pd.read_csv('Mall_Customers.csv')
print("Dataset shape:", df.shape)
df.head(8)
```

- Imported libraries for data handling (pandas, numpy), visualization (matplotlib, seaborn), and ML (scikit-learn).
- Loaded the dataset and displayed the first few rows.

➤ **Step 3 – Clean Column Names:**

```
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")
df = df.rename(columns={'customerid': 'customer_id', 'spending_score_1-100': 'spending_score'})
df.head()
```

- Standardized column names to lowercase with underscores for consistency.
- Renamed key columns for easier access.

➤ **Step 4 – Exploratory Data Analysis (EDA):**

```
df.describe()
sns.pairplot(df[['age', 'annual_income_k', 'spending_score']], diag_kind='kde')
```

- Generated statistical summary of the dataset.
- Visualized relationships among features to detect patterns.

➤ **Step 5 – Feature Selection:**

```
features = ['age', 'annual_income_k', 'spending_score']
X = df[features]
```

- Selected **Age, Annual Income, and Spending Score** as the primary features for clustering.
- These features are most relevant for identifying customer behavior patterns.

➤ **Step 6 – Data Scaling:**

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

- Standardized features to have mean = 0 and variance = 1.
- Ensures fair distance-based clustering, since raw income/spending values vary in scale.

➤ *Step 7 – Dimensionality Reduction with PCA:*

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
```

- Reduced dimensions to 2 for visualization while preserving maximum variance.
- PCA makes high-dimensional clustering interpretable in 2D scatterplots.

➤ *Step 8 – Elbow Method for Optimal Clusters:*

```
sse = []
for k in range(2, 10):
    km = KMeans(n_clusters=k, random_state=42)
    km.fit(X_scaled)
    sse.append(km.inertia_)

plt.plot(range(2,10), sse, marker='o')
plt.xlabel("Number of Clusters")
plt.ylabel("SSE")
plt.show()
```

- Calculated **Sum of Squared Errors (SSE)** for different cluster numbers.
- Used the **elbow point** to find the optimal number of clusters.

➤ *Step 9 – Silhouette Score Validation:*

```
for k in range(2,10):
    km = KMeans(n_clusters=k, random_state=42)
    labels = km.fit_predict(X_scaled)
    score = silhouette_score(X_scaled, labels)
    print(f"k={k}, silhouette={score:.3f}")
```

- Evaluated clustering quality for different k values.
- **Higher silhouette score = better-defined clusters.**

➤ *Step 10 – Apply K-Means Clustering:*

```
best_k = 4
kmeans = KMeans(n_clusters=best_k, random_state=42)
df['cluster'] = kmeans.fit_predict(X_scaled)
```

- Finalized **4 clusters** based on elbow + silhouette method.
- Assigned each customer to a cluster.

➤ *Step 11 – Cluster Visualization:*

```
plt.figure(figsize=(8,6))
sns.scatterplot(x=X_pca[:,0], y=X_pca[:,1], hue=df['cluster'], palette='Set1', s=80)
plt.title("Customer Clusters (PCA 2D Projection)")
plt.show()
```

- Visualized the clusters in **2D PCA space**.
- Different colours represent different customer groups.

➤ **Step 12 – Cluster Profiling:**

```
centroids = kmeans.cluster_centers_  
print("Cluster centroids (scaled features):", centroids)
```

- Examined cluster centers to understand characteristics of each group.
- Example: one cluster corresponds to “High Income – Low Spend.”

➤ **Step 13 – Human-Friendly Labels:**

```
label_map = {  
    0: "Medium Income / Medium Spend",  
    1: "Young High Spenders",  
    2: "Low Income / Low Spend",  
    3: "High Income / Low Spend"  
}  
df['cluster_label'] = df['cluster'].map(label_map)  
df[['customer_id', 'age', 'annual_income_k', 'spending_score', 'cluster_label']].head()
```

- Added descriptive labels for business interpretation.

➤ **Step 14 – Save Models and Results:**

```
joblib.dump(scaler, "scaler.pkl")  
joblib.dump(kmeans, "kmeans_model.pkl")  
df.to_csv("clustered_customers.csv", index=False)
```

- Saved the scaler, trained model, and final dataset for future use.

❖ **Results and Observations:**

➤ **5.1 Optimal Number of Clusters:**

- The **Elbow Method** showed a distinct bend at $k = 4$, suggesting that four clusters provide the most balanced segmentation.
- The **Silhouette Score** was also highest around $k = 4$, confirming that customers are best grouped into four segments.

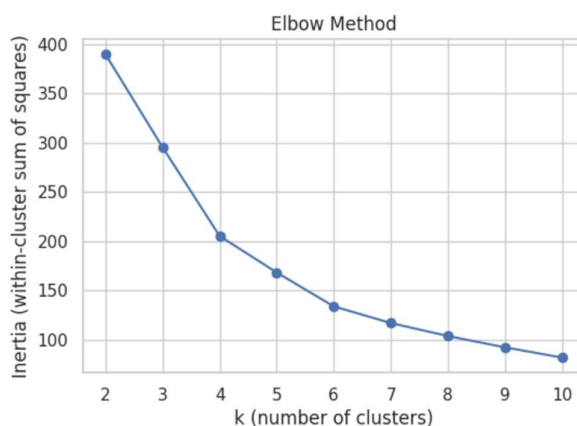


Figure 1: Elbow Method showing optimal $k = 4$

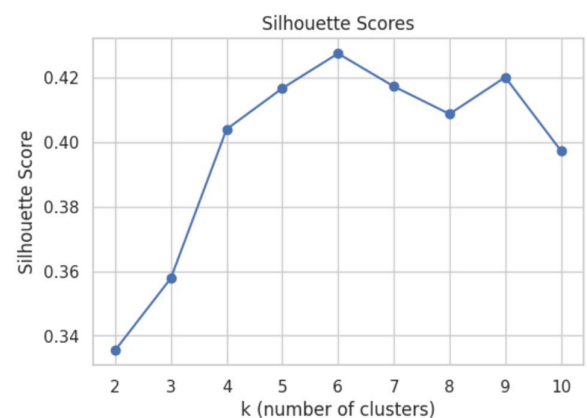


Figure 2: Silhouette Score analysis validating 4 as optimal cluster choice

➤ 5.2 Cluster Profiles:

Based on centroid analysis and visual inspection, the following customer segments were identified:

- **Cluster 0:** Medium Income – Medium Spend customers.
- **Cluster 1:** Young High Spenders → high potential for targeted marketing.
- **Cluster 2:** Low Income – Low Spend customers → least profitable segment.
- **Cluster 3:** High Income – Low Spend customers → potential luxury buyers but low engagement.

	age	annual_income_k	spending_score	cluster
0	41.94	88.94	16.97	0
1	56.33	54.27	49.07	1
2	25.25	25.83	76.92	2
3	32.69	86.54	82.13	3
4	26.68	57.58	47.79	4
5	45.52	26.29	19.38	5

Figure 3: Centroid table showing average values for Age, Annual Income, and Spending Score across clusters

➤ 5.3 Visualizations of Clusters:

Annual Income vs Spending Score Scatter Plot

- Displays clear separation of customer groups.
- Cluster centroids (black “X” markers) highlight typical customer behavior in each group.

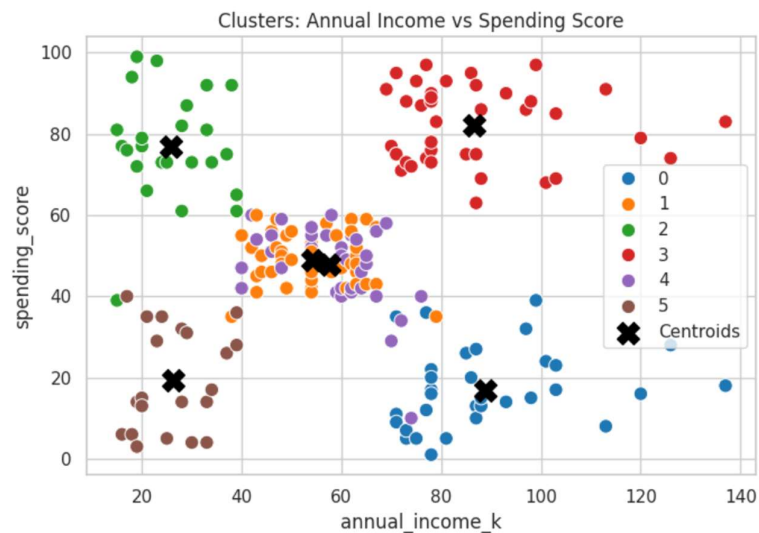


Figure 4: Scatter plot of customers by Annual Income vs. Spending Score

PCA 2D Plot

- Principal Component Analysis (PCA) reduced the data to two dimensions.
- Visualization shows well-separated clusters, validating the clustering results.

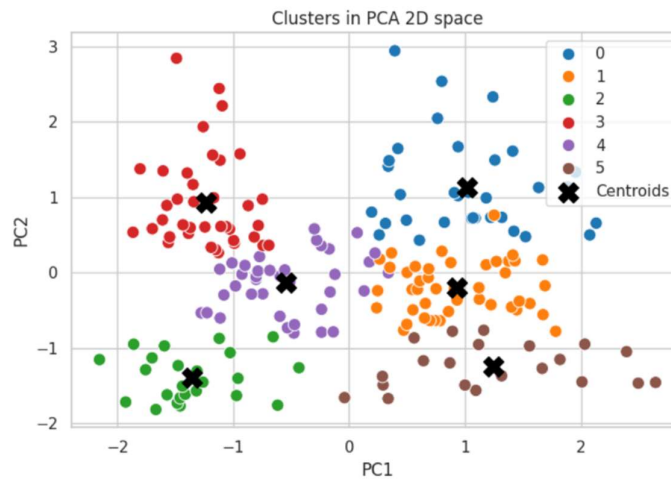


Figure 5: PCA visualization of clusters in 2D space.

❖ Conclusion:

The *Retail Store Customer Clustering* project successfully achieved its objective of segmenting customers based on their demographic and spending behavior using unsupervised learning techniques. By applying **K-Means Clustering** and **Principal Component Analysis (PCA)**, meaningful customer segments were identified and visualized clearly through scatter plots and PCA projections.

The analysis revealed **four distinct customer clusters**, each representing a unique purchasing pattern — from high-income, low-spending individuals to young, high-spending customers. These insights can help retail businesses **design targeted marketing strategies**, improve customer engagement, and enhance overall profitability.

Through this project, I gained practical understanding of how clustering algorithms work, how **dimensionality reduction** simplifies visualization, and how **data-driven insights** support business decision-making. The project aligns closely with the objective of applying **unsupervised machine learning** to real-world retail data and demonstrates the importance of combining statistical analysis with visualization for actionable outcomes.

❖ References:

- Kaggle Dataset: *Mall Customers Dataset*
- Scikit-learn Documentation: <https://scikit-learn.org/>
- Seaborn Documentation: <https://seaborn.pydata.org/>
- Dataset Link: <https://www.kaggle.com/datasets/shwetabh123/mall-customers>

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